

Korebami O. Adebajo

Mechanical Engineering Graduate Student

Education

2021 - 2025 **B.S.E**, Mechanical Engineering, University of South Carolina GPA: **3.9**

Minors in Electrical Engineering and Mathematics

Graduation with Leadership Distinction in Research

2025 - 2029 **Ph.D**, Mechanical Engineering, University of South Carolina

Research Experience

February 2022-present **Undergraduate Researcher**, Adaptive Real Time Systems Lab
Department of Mechanical Engineering, **University of South Carolina**

Multi-domain Modeling and Battery Routing of an Electric Airplane

- Developed and implemented a multi-domain model of an electric 6-seater Cessna 206 using MATLAB-Simscape to evaluate and optimize battery performance in electric aircraft, closely aligning with industry standards seen in the Pipistrel Velis Electro.
- Constructed simulations to analyze battery degradation based on different flight routes, load profiles, and State of Charge ranges. Evaluated the viability of specific SOC ranges in reducing battery degradation by up to 3%, contributing valuable insights into the long-term commercial feasibility of electric aviation.
- Created predictive models for Remaining Useful Life estimation, incorporating thermal and mechanical stress variables, to guide sustainable and economical battery usage in electric aircraft.
- Aimed to provide a scalable and environmentally focused framework for aviation electrification, with broader implications for sustainable transportation solutions across various sectors.

UAV Deployed Sensor Packaging

- Tuned and calibrated the control systems for a hexacopter to achieve stable flight and precise maneuvering during docking operations.
- Developing a deep learning model that replicates pilot actions using end-to-end control, enabling the UAV to autonomously identify, track, and navigate towards the docking station based on visual data.
- Applied advanced mathematical modeling techniques to implement autonomous decision-making capabilities for improved UAV-based sensor deployment on large civil structures.

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June - **Summer Research Intern** Therapeutic Technology Design and Development Lab
November - Department of Mechanical Engineering, **Massachusetts Institute of Technology**
2024

- Developed a MATLAB-based predictive model to characterize and optimize the decomposition of hydrogen peroxide in propellant-based actuators for untethered soft robotics, focusing on critical parameters such as chamber geometry, catalyst size, reaction rate, temperature, and pressure.
- Introduced the concept of residence time in the reactor, establishing its importance for fuel efficiency and control, with findings indicating that reducing the reactor's width-to-height ratio enhances residence time, resulting in improved predictability and power.
- Conducted experimental testing and data analysis to validate model predictions, adjusting key variables such as fuel volume, flow rate, and catalyst surface area to assess their impact on pressurization rates.
- Redesigned the fuel reactor engine to achieve a full-seal, eliminating leaks and improving retention by 50%, enabling reliable integration in full robotic systems.
- Presented preliminary results, demonstrating the model's potential to enhance actuator performance and opening possibilities for advanced applications in soft robotics.
- Mentored a high school student in a parallel project, guiding their analysis of fuel injection methods and fostering their understanding of energy-efficient designs in soft robotics.

Non-Research Related Experiences

January - **Mathematics Grader** Applied Linear Algebra and Elementary Differential Equations
December - Department of Arts and Sciences, University of South Carolina
2024

- Accurately and timely grade quizzes for approximately 100 students in both Math courses.
- Maintain clear communication with the course instructor regarding grading criteria and student progress.

August **Peer Tutor**
2022 - May Student Success Centre, University of South Carolina
2025

- Organizing developmental activities and sessions for 6-8 students weekly for various math, science, introductory engineering courses and Creo CAD software.
- Promoting and increasing inclusivity and diversity among over 200 USC students through services at the Student Success Center

- September 2023 - May 2024 **Formula 3 EV Tractive Team Lead**, Society of Automobile Engineers
University of South Carolina
- Implemented a regenerative braking system with required low and high-voltage circuits and displays for the race car.
 - Led a team of 7 students in the design and development of the tractive systems of the EV race car.
- September 2023 - May 2024 **Skills Robot Build Team Member**, Gamecock Robotics
University of South Carolina
- Collaborated with team members to improve the robot's chassis by integrating two piston-powered wings.
 - Participated in constructing skill robots based on project goals for the VEX Robotics competition.
- February - April 2024 **Student Participant**, Close Family Emerging Leaders Program
- Partook in team-building workshops on the significance of collaboration, empathy, and community involvement.
 - Engaged in a comprehensive curriculum focused on leadership, communication, and ethical decision-making.
- January - March 2024 **PwC X SGAP Extern**, Extern (previously known as Paragon One)
- Conducted comparative analysis identifying key areas for improvement and operational efficiency.
 - Presented strategic recommendations to NGO officials, showing strong communication skills.
- November 2022-Present **Pi Tau Sigma**, Mechanical Engineering Honor Society
- Collaborated with others in constructing the wooden Tiger for USC's annual Tiger burn using power saws and drills.
 - Coordinated with 6 members of the event planning committee for the Diversity Gala to connect students and 60 professionals at USC.

Publications and Posters

Peer Reviewed Conference Publications

- [3] Harshal Kaushik, Ali Mahboub Rad, **Korebami Adebajo**, Sobhan Badakhshan, Nathaniel Cooper, Austin Downey, Jie Zhang "Hybrid Powertrain Optimization for Regional Aircraft Integrating Hydrogen Fuel Cells and Aluminum Air Batteries" IDETC 2025
- [2] **Adebajo, O. Korebami**, Anthony, G., Peskar, J., Downey, A.R.J., Chen, Y., & Hu, C. "Impact of Charging Rate and State of Charge on Electric Aircraft Battery Degradation Using a Multi-Domain Model with Realistic Southeastern U.S. Flight Paths". AIAA SciTech Forum 2025.
- [1] Samuel Dutra Gollob, Alberto Comoretto, Qifan Yu, Matthew Fernandez, **Korebami O. Adebajo**,

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Johannes T. B. Overvelde, Kaitlyn Becker, Ellen Roche "An Actuator-Scale, Pressure-Amplifying, Monopropellant Engine for Untethered Pneumatic Systems". IEEE RoboSoft Conference 2025.

Conference Publications

- [1] Zheng Qi, Satme Joud N., **Adebajo O. Korebami**, Yount Ryan, & Downey Austin R.J. "Progress Towards Autonomous UAV Deployment of Sensor Package for Infrastructure Monitoring using Deep Learning Control". In preparation for SPIE Conference.

Poster Presentations

- [3] **Adebajo, O. Korebami**. Simulink Modeling of an Electric Airplane. Discover USC 2023 (Poster Presentation). **Discover USC Award for excellence**.
- [2] **Adebajo, O. Korebami**. Battery Degradation Prediction Aided by Multi-Domain Modeling. SC Battery Symposium, 2024 (Poster Presentation).
- [1] **Adebajo, O. Korebami**. Experimental Analysis of Catalyzed Hydrogen Peroxide Decomposition for Monopropellant-Powered Actuators. MIT MSRP Research Symposium 2024

Honors and Awards

Honors

- [4] **Discover USC Honourable Mention** "Multi-domain Modeling of an Electric Airplane" Poster Award for excellence and achievement 2023
- [3] **Capstone Scholar and Fellow** University of South Carolina, 2021-Present
- [2] **Dean's List** University of South Carolina, GPA > 3.5, Fall 2022 - Present
- [1] **President's List** University of South Carolina, 4.0 GPA list, Fall 2021 and Spring 2022

Scholarships

- [4] **Colorstack Funding** \$1500 academic scholarship awarded in 2024
- [3] **Boeing Scholarship** \$1500 academic scholarship awarded in 2024
- [2] **Herty B. Mechanical Engineering Scholarship** \$1000 academic scholarship awarded in 2023
- [1] **Academic Scholar Distinction Award** \$2000 academic scholarship awarded in 2021

Research Grants

- [4] **Magellan Scholar Award** "Multi-domain Modeling of an Electric Airplane" \$2500 research funding awarded in April 2024
- [3] **Michael A. and Elizabeth Ann Sutton Fellowship** "Multi-domain Modeling of an Electric Airplane" \$1400 research funding awarded in February 2023
- [2] **Magellan Apprentice Grant** "Simulink Modeling of an Electric Airplane" \$1000 research funding awarded in March 2023
- [1] **Magellan Journey Grant** "Simulink Modeling of an Electric Airplane" \$1000 research funding awarded in July 2022